In the Module 11 Lab, my primary contribution was writing, executing, and analyzing the hyperparameter tuning workflows and the AutoML pipeline in Google Colab, as well as completing the final evaluation and committing the finished notebook to my GitHub repository.

Baseline Setup & Data Preparation

I loaded the Iris dataset using scikit-learn, split the features and target labels into a 70/30 training and testing split (random_state=42), and trained a baseline RandomForestClassifier with default hyperparameters that reached 100.00% test accuracy.

Task 1: Grid Search Implementation

I built the param_grid dictionary to test combinations of n_estimators (50, 100, 200) and max_depth (5, 10, None).

I configured and fit GridSearchCV with 5-fold cross-validation (cv=5, n_jobs=-1), running 45 total model fits to identify the best parameters (max_depth: 5, n_estimators: 100) with a 94.29% cross-validated accuracy score.

Task 2: Random Search Implementation

I expanded the search space by creating param_dist using np.linspace(50, 500, 10) for n_estimators, five options for max_depth (5, 10, 20, 30, None), and three options for min_samples_split (2, 5, 10).

I set up and executed RandomizedSearchCV across 10 iterations (n_iter=10, cv=5, random_state=42) for 50 total fits, finding an optimal configuration (n_estimators: 200, min_samples_split: 5, max_depth: 20) that matched the 94.29% cross-validated score while covering a much broader parameter range.

AutoML Execution & Knowledge Check Analysis

I installed AutoGluon in the Colab environment, formatted the training and testing sets into combined pandas DataFrames with the species target column, and ran TabularPredictor with a 60-second time limit to train 11 level-1 models and a WeightedEnsemble_L2 model.

I evaluated the resulting leaderboard—comparing validation vs. test scores across models like XGBoost, RandomForest, ExtraTrees, LightGBM, and CatBoost—and wrote out the Knowledge Check responses explaining the difference between model parameters and hyperparameters, when to use Grid Search versus Random Search, and how AutoML automates the end-to-end machine learning workflow.